

Label Sensitivity of Exact Subgroup First Passage

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Abstract

We resolve the labelwise sensitivity of the exact random-order first-passage variables from the frozen sixteen-label closure model. For each of twenty subgroup targets and all sixteen labels, we enumerate the exact pivotal-rank law over all $16!$ permutations. The pivotal probabilities, rank-weighted contributions, and rank-square contributions recover the C88 first-passage mass, mean, and second moment by exact efficiency identities. The complete 20×16 atlas is independently reconstructed and checked symbolically. This is a finite combinatorial result under `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Definition

Let $L = \{S_1, \dots, S_{16}\}$ and let T_H be the C88 first prefix time whose generated subgroup contains H . A label S_i is pivotal at rank k when it belongs to the hitting prefix A_k but deleting it from that prefix removes the target containment. Write $p_H(i, k)$ for the number of permutations with this event and first passage at k .

2 Exact identity

Theorem 1. *For every target H , label S_i , and $k \geq 1$,*

$$p_H(i, k) = \#\{A : |A| = k, S_i \in A, H \leq \Phi(A), H \not\leq \Phi(A \setminus \{S_i\})\} (k-1)!(16-k)!.$$

For every nontrivial target,

$$\sum_{i,k} p_H(i, k) = 16!, \quad \sum_{i,k} k p_H(i, k) = 16! \mathbb{E}[T_H],$$

and the analogous identity with k^2 recovers $16! \mathbb{E}[T_H^2]$. For the trivial target all terms vanish.

Proof. The displayed support count fixes the unordered prefix and its last label. The first $k-1$ positions and the suffix can then be ordered independently, giving the factorial factor. Each nontrivial permutation has exactly one last pivotal label at its first hitting prefix, so summing over labels and ranks partitions the first-passage permutations. Multiplication by k or k^2 gives the moment identities. $\square$

3 Complete finite atlas

The canonical receipt contains all 320 target-label rows, every rank cell, reduced pivotal probability, conditional pivotal rank mean, and the global label totals. The top target has mean 36499/3960 as in C83/C88. The order two target is pivotal only through S_9 . The top target has ten nonzero label contributions, all recorded in the receipt; these are observations of the frozen named coordinates, not an affine or arithmetic classification.

4 Audit and scope

An independent checker reconstructs every pivotal row directly from the C88 hit bitsets. SymPy verifies the probability, first-moment, second-moment, and generating-polynomial identities. Clean replay preserves the evidence hash

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No arithmetic/local data, Euler factors, root numbers, automorphy, full Burnside ring or table of marks, or Hilbert–Polya operator is claimed.